$k = 4$ index calculus in characteristic 2: Frobenius-reduced $\mathbf{D_4}$ system, C_4 481× lower, IC ÷ rho falling as $n^{-1/4}$ (crossover extrapolated near 2^{68})

2026-10-15. `research/gaudry_quartic_d4_char2_20261015/`. Protocol registered (commit `666438375`) before any claim data; ecbench binary `92610bdf…` throughout. **Class: engineering:** a constant in one decomposition, plus a measured trend; the exponent is unchanged and no run reaches the crossover. Every run is listed in `EXPERIMENTS.md`.

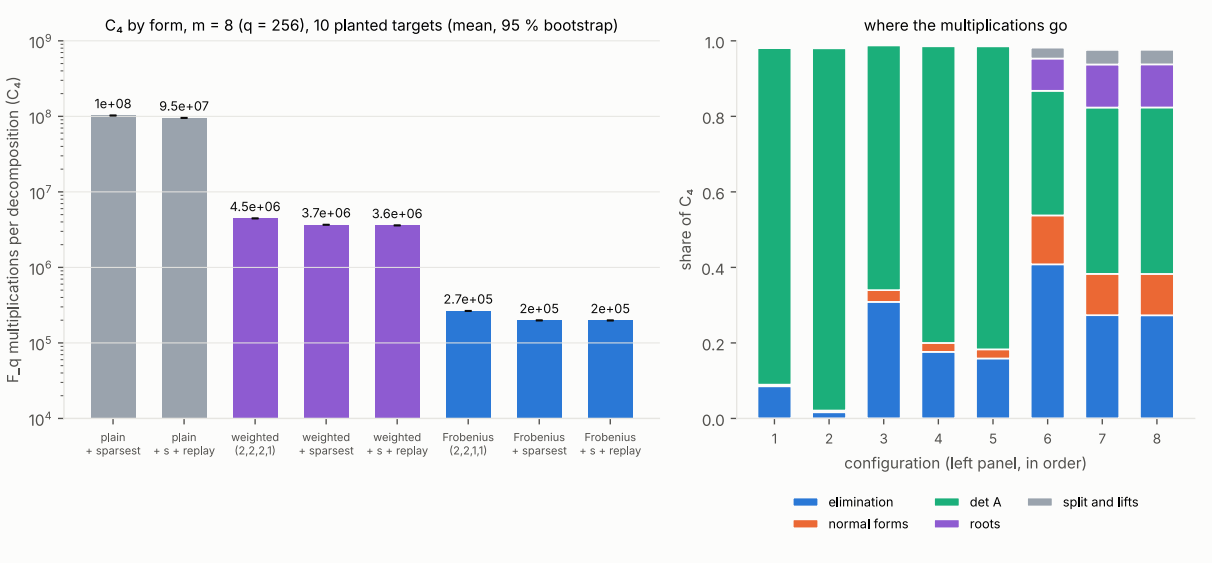
The system

With $T = \left(0, \sqrt{b}\right)$ and $t = \beta/(x+\beta)$, $\beta = b^{1/4}$, adding T is $t \mapsto t+1$. The invariants are $w = t^2 + t$, $s_k = e_k(w)$ and $\pi = \sum t$ with $\pi^2 + \pi = s_1$. Only 21 of 105 monomials carry the interpolated relation, all even in s_2, s_3, π, so the system is rewritten in $(s_2^2, s_3^2, s_4, \pi^2)$: weights $(2,2,1,1)$, 64 solutions, Macaulay degree 11, a 384×448 matrix (plain: 11116×11628). Chains, sparsest pivots and replay with top-up come from the odd study (1863).

One decomposition (C_4, $\mathbb{F}_q$ multiplications; run report `run-20261015T0530Z-ladder`)

configuration ($m = 8$, 10 planted)	matrix	C_4
plain, degree 14, + sparsest + replay	11116×11628	$9.54 \cdot 10^7$
weighted (2,2,2,1), degree 26, + s + r	3688×4200	$3.60 \cdot 10^6$
Frobenius (2,2,1,1), degree 11, + s + r	384×448	$\mathbf{1.98 \cdot 10^5}$

Paired ratios (95 % bootstrap): plain/weighted 26.5× (26.4–26.6), weighted/Frobenius 18.1× (18.08–18.18), **plain/Frobenius 481×** (479–483). Across $m = 5..15$ the Frobenius C_4 is $1.13–2.50 \cdot 10^5$; only root finding grows. 512 closed solves, none disagreeing with MITM; the 8 planted failures ($m = 5, 6$) are positive-dimensional families (every decomposition contains $\{t, t+1\}$), and no uniform target fails.



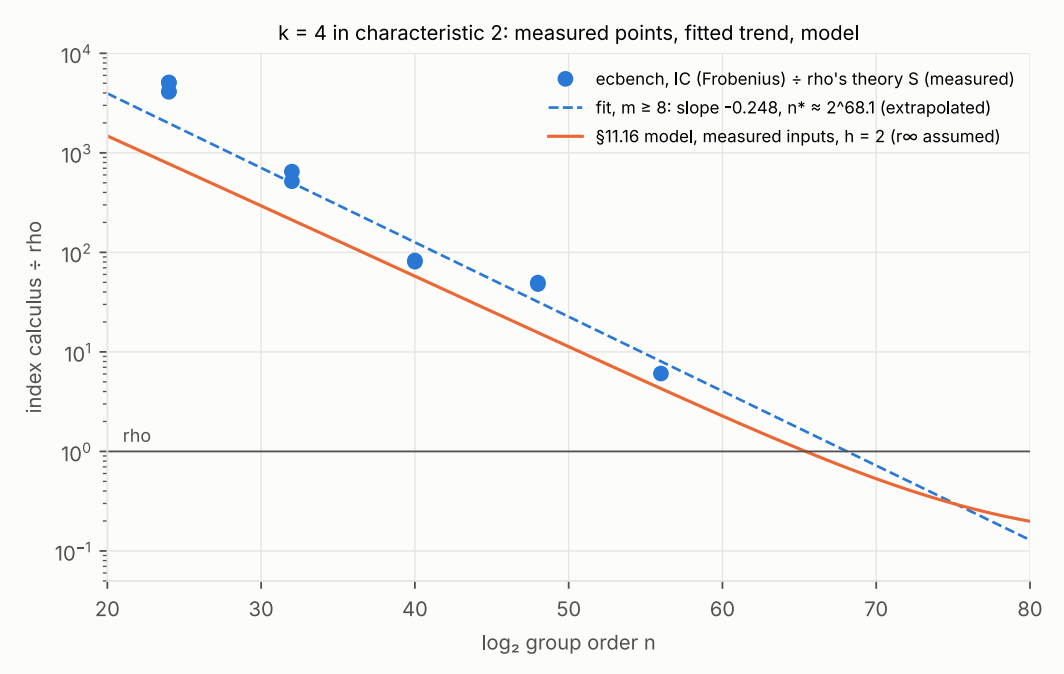
C_4 by option at $m = 8$ and its split by phase.

End to end (ecbench; S = group-addition equivalents ÷ $\sqrt{r}$)

At $m = 6$ (session `ECBS1h03a1d62aba88`, 10/10 verified) the three forms use identical trials and relations; S is 713,704 / 722,233 (plain), 38,542 / 31,128 (weighted), **4,501 / 3,654 (Frobenius)**, rho 1.017 / 1.314. `ecbench compare`: Frobenius/plain 0.0057 (0.0051–0.0063).

m	$\log_2 n$	h	IC (Frobenius) S	IC ÷ rho theory
6	24	14	4,501; 3,654	5,080; 4,124
8	32	6	573.9; 459.6	648; 519
10	40	2	71.4; 73.4	80.6; 82.9
12	48	10	44.0; 42.6	49.7; 48.1
14	56	2	5.37 (one target)	**6.06**

Sessions `ECBS1h4d63e1551dc1` (24/24) and `ECBS1hb9229e3a0f30` (3/3). IC is still 6.1× rho's theory (4.3× rho's measured run) at $n \approx 2^{56}$. The fit of $\log_2(S_{\mathrm{IC}}/0.886)$ against $\log_2 n$ for $m \geq 8$ has slope **−0.248** ($n^{-1/4}$ expected) and reaches rho at $\boldsymbol{n^* \approx 2^{68.1}}$ (leave-one-out $2^{66.7}$–$2^{71.6}$): an extrapolation. The model with measured inputs gives $2^{63.1}$–$2^{69.0}$ for $h = 1, 2, 4$ (r_∞ assumed). The uneven steps follow the cofactor h (the ratio carries $\sqrt{h}$); this reading is post hoc.



IC S ÷ rho's theory S per run, the fit ($m \geq 8$) and the model. `raw/summary.json`.

Predictions

A1 (exact), A2 ($C_4 \in [8 \cdot 10^4, 4 \cdot 10^5]$), A3 (closing), A4 (ablation), B1 (verified, replays identical, equal trials, order), B2 (ratio ranges at $m = 8, 10, 12, 14$), C1 (slope in $[-0.35, -0.15]$) and C2 (n^* in $[2^{60}, 2^{85}]$): **all held** (B1′s replay part pending). The replay audit (`verify --replay-all`) reproduced all 24 runs of B2; those of B1 and B3 are running.

Scope. Curves over $\mathbb{F}_{2^{4m}}$ are not standard curves; the crossover is an extrapolation plus a model, and $m = 14$ is one target. Prime-degree binary fields are settled separately (1885).

Environment: Claude Code cloud container, Xeon @ 2.10 GHz, 4 vCPU, one core reserved per session. Conductor task T-2.